

K-Means Clustering: Categorizing Attack Patterns in Unlabeled Phishing URLs

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Overview

This NB applies K-Means clustering to all tuples labeled "other" from the PhishTank dataset; these are URLs that could not be attributed to a specific target. Our goal is to find whether these "others" could be classified by meaningful patterns based on their URL structures alone. "Others" account for ~90% of the Phishtank data points. We utilize the very same lexical features from the SVM notebook (2.0) but apply unsupervised learning instead.

✓ Import dataset from github

```
import os

if not os.path.exists('DS_MLProject_ColabIntegration'):
    !git clone https://github.com/ns15468-gasou/DS_ML_Project_ColabIntegration.git
os.chdir('DS_ML_Project_ColabIntegration/Project/notebooks')
print("Working directory", os.getcwd())
```

```
fatal: destination path 'DS_ML_Project_ColabIntegration' already exists and is not an empty
Working directory /home/nic/DS_ML_Project_ColabIntegration/Project/notebooks/DS_ML_Project
```

✓ Imports

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
import numpy as np
from urllib.parse import urlparse
```

✓ Data Loading

Here we filter to only data points labeled as "other" for Target.

```
phish = pd.read_csv('../data/processed_input/phishtank_cleaned.csv')

other = phish[phish['target']=='Other'].copy()
print(f"total Phishtank rows: {len(phish)}")
print(f"'Other' rows: {len(other)}")
other.head()
```

```
total Phishtank rows: 56045
'Other' rows: 52773
```

	phish_id	url	phish_detail_url	submission_time
0	9358496	https://sgpass-fr033.com/	http://www.phishtank.com/phish_detail.php? phis...	2026-03-03 19:50:49+00:00
1	9358495	https://sgpass-fr033.com/pages/asset.php	http://www.phishtank.com/phish_detail.php? phis...	2026-03-03 19:50:36+00:00
2	9358485	https://dervi-emprendec.webcindario.com/	http://www.phishtank.com/phish_detail.php? phis...	2026-03-03 19:27:05+00:00
6	9358476	https://passage-route-mobilites.com/	http://www.phishtank.com/phish_detail.php? phis...	2026-03-03 19:16:14+00:00
7	9358475	https://www.passage-route-mobilites.com/	http://www.phishtank.com/phish_detail.php? phis...	2026-03-03 19:16:04+00:00

5 rows × 21 columns

✓ Feature Engineering

We divide the URLs into 6 features as with NB2.0

```
phish['host'] = phish['url'].apply(lambda x: urlparse(x).netloc.lower())
phish['full_length'] = phish['url'].str.len()
phish['host_length'] = phish['host'].str.len()
phish['path'] = phish['url'].apply(lambda x: urlparse(x).path)
phish['path_depth'] = phish['path'].apply(lambda p:
p.strip('/').count('/') + 1 if p.strip('/') else 0)
phish['dot_count'] = phish['url'].str.count(r'\.')
phish['hyphen_count'] = phish['url'].str.count('-')
phish['digit_count'] = phish['url'].str.count(r'\d')
phish['digit_ratio'] = phish['digit_count'] / phish['full_length']
phish['subdomain_depth'] = phish['host'].str.count(r'\.')
```

```
other = phish[phish['target'] == 'Other'].copy()
print(other.shape)
```

```
(52773, 21)
```

✓ Scaling

K-Means is distance based, so we want our features to be on the same scale, thus we use StandardScaler

```
features = ['host_length', 'path_depth', 'dot_count', 'hyphen_count', 'digit_ratio', 'sul

X = other[features].dropna()
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print(f"Feature matrix shape: {X_scaled.shape}")
```

```
Feature matrix shape: (52773, 6)
```

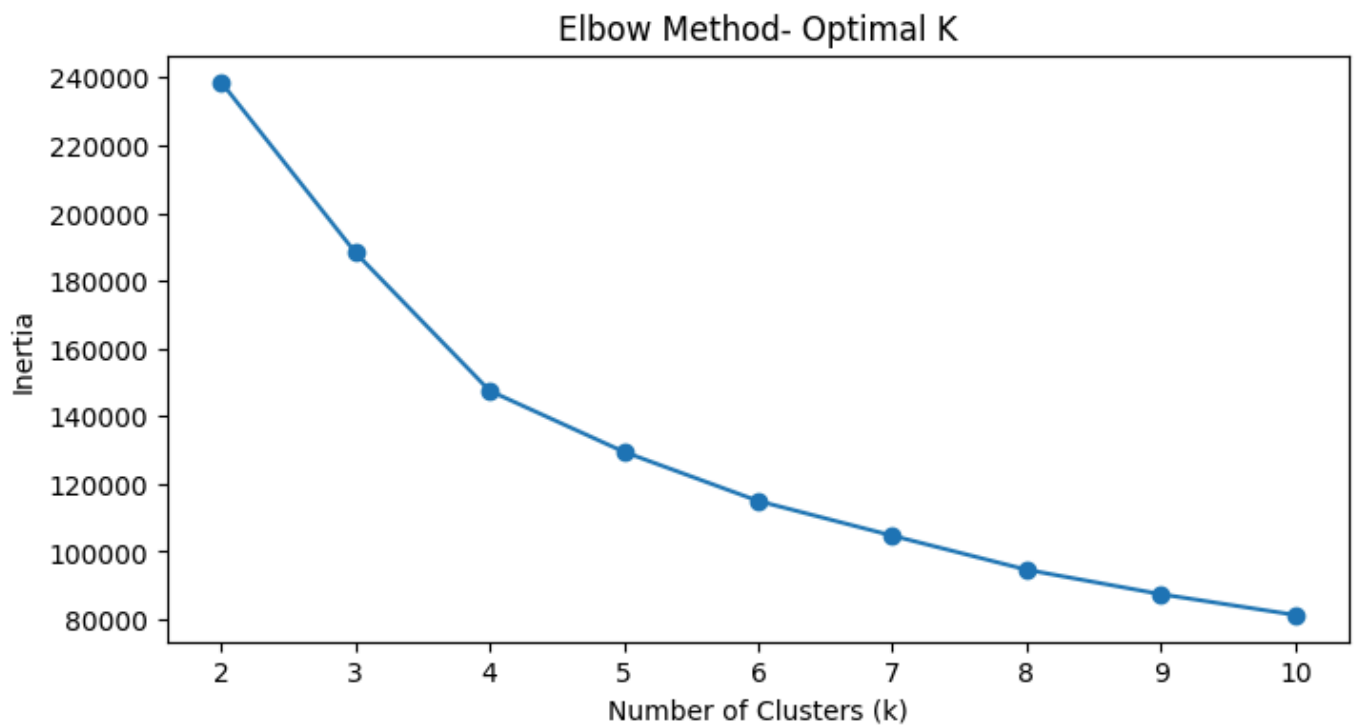
✓ Elbow Method

This cell graphically represents how and why we have chosen 4 groups.

```
inertia = []
k_range = range(2,11)

for k in k_range:
    km = KMeans(n_clusters=k, random_state=777, n_init=10)
    km.fit(X_scaled)
    inertia.append(km.inertia_)

plt.figure(figsize=(8,4))
plt.plot(k_range, inertia, marker='o')
plt.title('Elbow Method- Optimal K')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.tight_layout
plt.show()
```



✓ Note

Seeing as the curve is nearly linear, there does not exist an extremely obvious choice for number of clusters. That said, the most drastic point of change in derivative is at $k=4$.

```
km = KMeans(n_clusters=4, random_state=777, n_init=10)
other['cluster'] = km.fit_predict(X_scaled)

print(other['cluster'].value_counts())
```

```
cluster
1    21341
3    15517
2     8680
0     7235
Name: count, dtype: int64
```

```
cluster_summary = other.groupby('cluster')[features].mean().round(2)
print(cluster_summary)
```

```
      host_length  path_depth  dot_count  hyphen_count  digit_ratio  \
cluster
0           38.65         0.59         2.82           1.09          0.24
1           21.29         0.67         2.15           0.44          0.03
2           15.88         4.51         2.40           2.11          0.10
3            8.75         1.07         1.07           0.34          0.04

      subdomain_depth
cluster
0                2.27
1                2.07
```

```
2          1.99
3          1.00
```

```
for c in range(4):
    print(f"\n--- Cluster {c} ---")
    print(other[other['cluster']== c]['url'].sample(5, random_state=777).values)
```

```
--- Cluster 0 ---
```

```
<StringArray>
```

```
['https://pub-7497b4cc97ab46a6a9a492eb1b6a38ec.r2.dev/index.html',
 'https://eu.jotform.com/app/251023382704347',
 'https://eu.jotform.com/app/251953383222355',
 'http://full-language-233771.framer.app',
 'https://soulful-time-409544.framer.app/']
```

```
Length: 5, dtype: str
```

```
--- Cluster 1 ---
```

```
<StringArray>
```

```
['https://sites.google.com/view/oiuiruieor98490krejjkljklejkef/home',
 'https://webkkwel.weebly.com/',
 'https://adminhelpdeskunit1.weebly.com/',
 'https://repliesemail80.wixsite.com/my-site-8',
 'https://id-ionos-de-webmail-system.blogspot.com/']
```

```
Length: 5, dtype: str
```

```
--- Cluster 2 ---
```

```
<StringArray>
```

```
['https://www.dropbox.com/scl/fi/rqwa7nyayqn676cdeav2d/BT-V0ICE.paper?dl=0&oref=e&r=AC0ESs']
```

```
Length: 5, dtype: str
```

```
--- Cluster 3 ---
```

```
<StringArray>
```

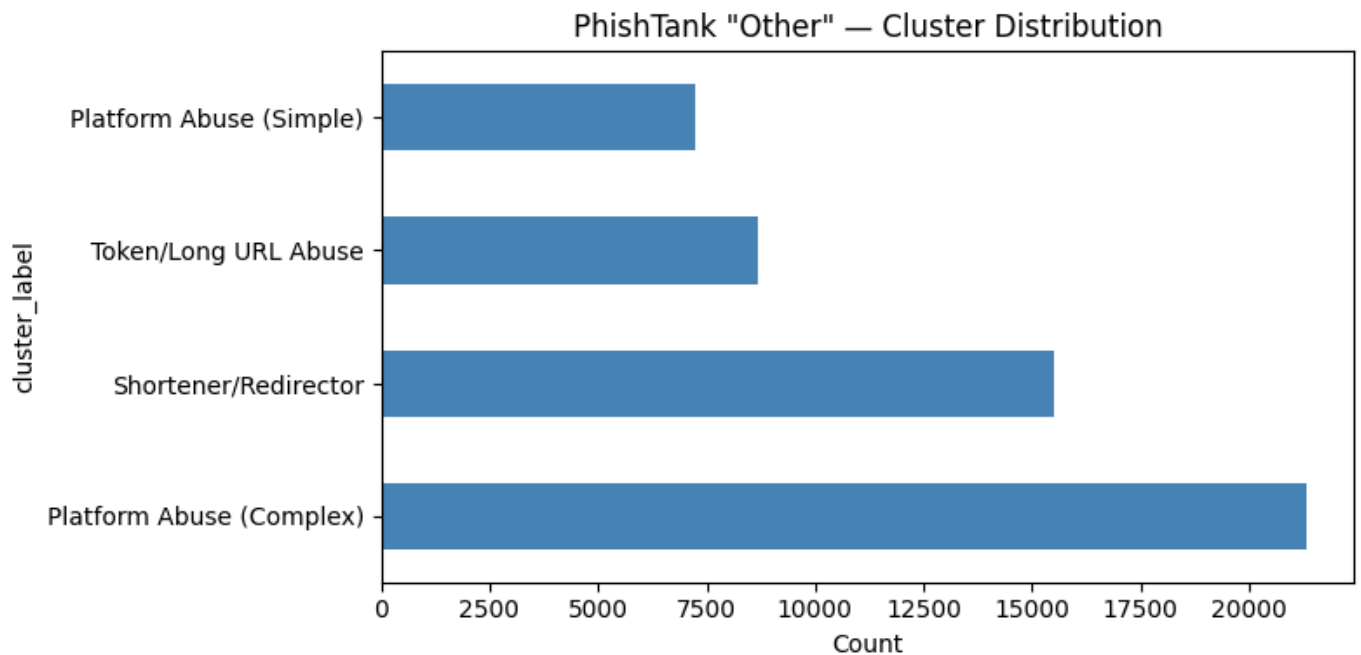
```
['https://qrco.de/bfyD0b',
 'https://t.co/4EOHuT9IeH',
 'https://fidelitysafetyfi.one/page.html',
 'https://rakutee.com/vpngzifq/',
 'https://qrco.de/bf0BG2']
```

```
Length: 5, dtype: str
```

```
cluster_names = {0: 'Platform Abuse (Simple)',
                  1: 'Platform Abuse (Complex)',
                  2: 'Token/Long URL Abuse', 3: 'Shortener/Redirector'}
```

```
other['cluster_label'] = other['cluster'].map(cluster_names)
other['cluster_label'].value_counts().plot(kind='barh', figsize=(8,4),
                                           color='steelblue')
```

```
plt.title('PhishTank "Other" – Cluster Distribution')
plt.xlabel('Count')
plt.tight_layout()
plt.show()
```



```
samples = []
for c in range(4):
    label = cluster_names[c]
    urls = other[other['cluster'] == c]['url'].sample(3,
random_state=42).values
    for url in urls:
        samples.append({'Cluster': label, 'Example URL': url[:60] +
'...' if len(url) > 60 else url})

sample_df = pd.DataFrame(samples)

fig, ax = plt.subplots(figsize=(12, 5))
ax.axis('off')
table = ax.table(cellText=sample_df.values, colLabels=sample_df.columns,

loc='center', cellLoc='left')
table.auto_set_font_size(False)
table.set_fontsize(8)
table.auto_set_column_width(col=list(range(len(sample_df.columns))))
plt.title('Sample URLs by Cluster', pad=20)
fig.patch.set_visible(False)
plt.savefig('cluster_samples.png', dpi=150, bbox_inches='tight',
transparent=True)
plt.show()
```

Sample URLs by Cluster

Cluster	Example URL
Platform Abuse (Simple)	https://pub-d637c8d427db484db2245748126bd788.r2.dev/rcubic.h...
Platform Abuse (Simple)	https://reserved-team-373417.framer.app/
Platform Abuse (Simple)	https://pub-cec2c769e7424225875847093b63e672.r2.dev/index.ht...
Platform Abuse (Complex)	https://please-wait--connection.firebaseio.com/
Platform Abuse (Complex)	https://wxooontreonr.blogspot.com/
Platform Abuse (Complex)	https://startus-home.ghost.io/live-enus/
Token/Long URL Abuse	https://docs.google.com/presentation/d/e/2PACX-1vSd2kgD9d1Ub...
Token/Long URL Abuse	http://zuza.stefek.szczecin.pl/Webmail/75/Webmail/webmail.ph...
Token/Long URL Abuse	https://docs.google.com/presentation/d/e/2PACX-1vTMeGlpqWyMI...
Shortener/Redirector	https://qrco.de/bgbcV6
Shortener/Redirector	https://i8.ae/cHKFq
Shortener/Redirector	https://tinyurl.com/3xht4cub

✓ Gradient Boosting Cluster Classifier

Uses K-Means cluster assignments as labels to train a supervised GBT classifier. Evaluated on an 80/20 split, then retrained on the full dataset.

```
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import classification_report, accuracy_score
from sklearn.model_selection import train_test_split

cluster_labels = km.labels_
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(
    X_scaled, cluster_labels, test_size=0.2, random_state=777
)

gbt_eval = GradientBoostingClassifier(n_estimators=100, random_state=777)
gbt_eval.fit(X_train_c, y_train_c)
y_pred_c = gbt_eval.predict(X_test_c)

print(f"Accuracy: {accuracy_score(y_test_c, y_pred_c):.4f}")
print(classification_report(y_test_c, y_pred_c, target_names=[cluster_names[i] for i in range(cluster_labels.max() + 1)]))

gbt_cluster_model = GradientBoostingClassifier(n_estimators=100, random_state=777)
gbt_cluster_model.fit(X_scaled, cluster_labels)
print("GBT cluster model trained on full dataset.")
```

Accuracy: 0.9945

precision recall f1-score support

Platform Abuse (Simple)	0.99	0.99	0.99	1490
Platform Abuse (Complex)	0.99	1.00	1.00	4258
Token/Long URL Abuse	0.99	0.99	0.99	1723
Shortener/Redirector	1.00	1.00	1.00	3084
accuracy			0.99	10555
macro avg	0.99	0.99	0.99	10555
weighted avg	0.99	0.99	0.99	10555

GBT cluster model trained on full dataset.

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